

```
praveenkumar@gmail:~  
GNU nano 8.7.1 Exp28.cpp  
#include <system.h>  
  
SC_MODULE(Counting_semaphore) {  
  
    int sem;  
  
    void task() {  
        wait(true); {  
            wait(1, SC_SEC);  
  
            if(sem > 0) {  
                sem--;  
                cout << "Task Accessing Resource at "  
                    << sc_time_stamp()  
                    << " Remaining: " << sem << endl;  
  
                wait(2, SC_SEC);  
                sem++;  
            }  
        }  
    }  
  
    SC_CTOR(Counting_semaphore) {  
        sem = 2;  
        SC_THREAD(task);  
        SC_THREAD(task);  
    }  
}
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo M-A Set Mark
M-G Copy

```
praveenkumar@gmail:~$ nano Exp28.cpp  
praveenkumar@gmail:~$ g++ -o Exp28 Exp28.cpp -lsystemc  
praveenkumar@gmail:~$ ./Exp28  
  
SystemC 3.0.2-Accellera +: Nov 5 2025 12:44:46  
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ALL RIGHTS RESERVED  
  
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_0  
In file: kernel/sc_object_manager.cpp:153  
Warning: (W505) object already exists: Counting.task. Latter declaration will be renamed to Counting.task_1  
In file: kernel/sc_object_manager.cpp:153  
  
Task Accessing Resource at 1 s Remaining: 1  
Task Accessing Resource at 1 s Remaining: 0  
Task Accessing Resource at 3 s Remaining: 0  
Task Accessing Resource at 4 s Remaining: 0  
Task Accessing Resource at 5 s Remaining: 0  
Task Accessing Resource at 6 s Remaining: 0  
Task Accessing Resource at 7 s Remaining: 0  
Task Accessing Resource at 8 s Remaining: 0  
Task Accessing Resource at 9 s Remaining: 0  
praveenkumar@gmail:~$ |
```